

CS 215: Task 02 Due: 15-Aug Time: as mentioned in Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read the data correctly, we might be able to.
 - You are highly encouraged to NOT use an LLM to solve this worksheet. If you get stuck, and want to use an LLM, make sure you tell the LLM that you don't want a solution, but an answer to your question. Don't say "give me a hint". Ask: "I am trying this problem in this way, am I right? Answer yes or no".
 - The problems build upon concepts, so it is recommended you approach them linearly. It is recommended that you do not attempt the questions out of order.
 - For some problems, check out the hints carefully.
 - For brevity, we will use probability density function as density function or p.d.f., or simply pdf.
 - In all cases, when you write an answer, give reasons. The space available is indicative of the length of the reason desired.
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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X is a continuous random variable with pdf

$$f(x) = \begin{cases} C(4x - 2x^2), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) C is

(b) $\mathbb{P}\{X > 1\}$ is

2. The amount of time, in hours, that a computer functions before breaking down is a continuous random variable with pdf

$$f(x) = \begin{cases} \lambda e^{-x/100}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- (a) What is the probability that a computer will function between 50 and 150 hours before breaking down?

- (b) What is the probability that it will function less than 100 hours?

3. The joint pdf of X and Y is given by

$$f(x, y) = \frac{6}{7} \left(x^2 + \frac{xy}{2} \right), \quad 0 < x < 1, \quad 0 < y < 2.$$

- (a) Verify that this is indeed a joint density function.

- (b) Compute the density function of X .

(c) $\mathbb{P}\{X > Y\}$ is

4. X and Y are independent continuous random variables. Show that

(a)

$$P\{X + Y \leq a\} = \int_{-\infty}^{\infty} F_X(a - y) f_Y(y) dy$$

where f_Y is the density function of Y , and F_X is the distribution function of X .

(b)

$$P\{X \leq Y\} = \int_{-\infty}^{\infty} F_X(y) f_Y(y) dy$$

where f_Y is the density function of Y , and F_X is the distribution function of X .

5. In a certain town, there exist 100 buses out of which 1 is red and 99 are blue. XYZ observes a serious accident caused by a bus at night and remembers that the bus was red in color. Hence, the police arrest the driver of the red bus. The driver pleads innocence. Now, a benevolent lawyer decides to defend the distressed bus driver in court. The lawyer ropes in an ophthalmologist to test XYZ's ability to differentiate between the colors red and blue, under illumination conditions similar to those that existed that fateful night. The ophthalmologist's expert deposition to the court is that XYZ sees red objects as red 99% of the time and blue objects as red 2% of the time.

If you are the defense lawyer how will you argue? That is, compute the probability that the bus was red, when XYZ thought it to be red, possibly correctly, possibly incorrectly. Provide reasons.

Teaching Team Notes:

- (a) In problems like this, it is good to start with something like: Let X be the event with such-and-such property. We are not sure about the outcome, so we assign a probability to this event. Similarly build other events, or random variables as the case may be, depending on how you like to think about events and random variables.
- (b) In this particular case, consider conditional probability given the wording of the problem statement.

6. $X_1, X_2, \dots, X_n$ are independent random variables each having the common (canonical) uniform density function

$$f(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Teaching Team Notes:

- (a) In a problem like this which may seem out of reach, first restrict the variables to $n=2$ and see if you can reason.
- (b) When we need the expectation, we need the pdf. For continuous random variables, consider looking at the distribution function and then differentiating it to get the pdf.
- (c) Since the problem specification talks of independent, the hint is to think of the joint pdf and then use the product formula.

$E[\text{Max}(X_1, \dots, X_n)]$ is

7. A community consists of 100 married couples. If 50 members of the community die, what is the expected number of marriages that remain intact? Assume that the set of people who die is equally likely to be any of the $\binom{200}{50}$ groups of size 50.

Teaching Team Notes:

(a) In problems like this, you are looking for using the linearity of functions along with indicator variables.

(b) For this problem, this indicator variable is relevant

$$X_i = \begin{cases} 1, & \text{if neither member of couple } i \text{ dies,} \\ 0, & \text{otherwise.} \end{cases}$$

(c) Also pay attention to when the marriage is intact, and when a person dies.